Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829 · LINE solaress1
www.solaress.org · www.โซลาร์เอสส์.com

TECHNICAL NOTE · BACK-CONTACT PANELS

RAPD 670W (BC) vs AIKO 665W (ABC): two back-contact panels compared

Specifications, inverter matching, heat, warranty and what to check before buying · Solar ESS · Rev 01 · 30 October 2026

24.8 / 24.6%
RAPD / AIKO module efficiency, worked out from power and the 2,382 × 1,134 mm size

16.2 / 14.6 A
RAPD / AIKO current at maximum power: the difference that matters most for the inverter

11 / 10
RAPD / AIKO panels per string on a 600 V inverter at 10 °C

15 yr
AIKO product warranty on the datasheet (12 years quoted by some Thai dealers)

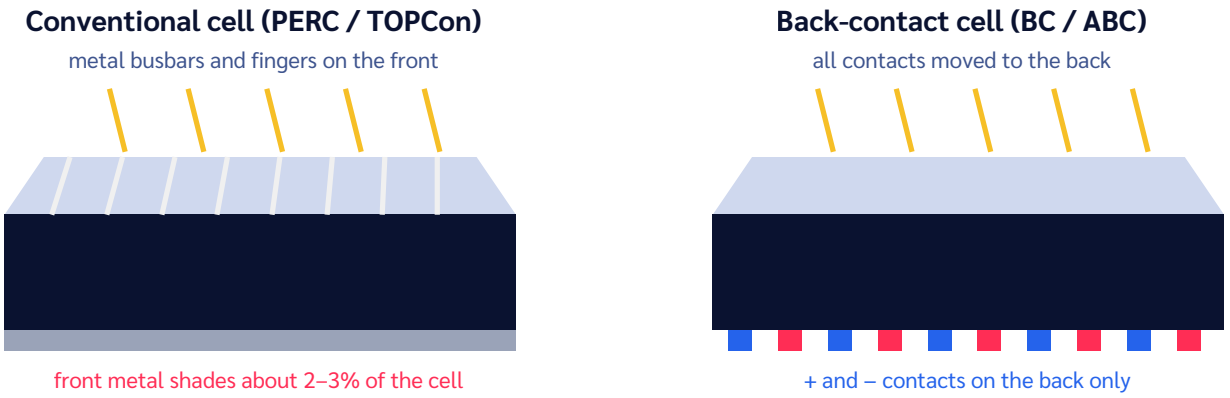
The short answer

Almost the same power
Both are 2,382 × 1,134 mm back-contact panels with no front busbars. 670 W against 665 W is less than 1% apart.

Different electrical design
RAPD is a high-current, lower-voltage panel (16.2 A, 41.4 V). AIKO is lower-current, higher-voltage (14.6 A, 45.6 V). Choose the one your inverter's MPPT inputs suit.

Paperwork decides
AIKO's figures come from its published datasheet. We could not find a manufacturer datasheet for the RAPD 670 W BC, so get it, with certificates, before you buy.

1 What back-contact (BC / ABC) means



Sunlight (yellow) reaches the whole front of a back-contact cell

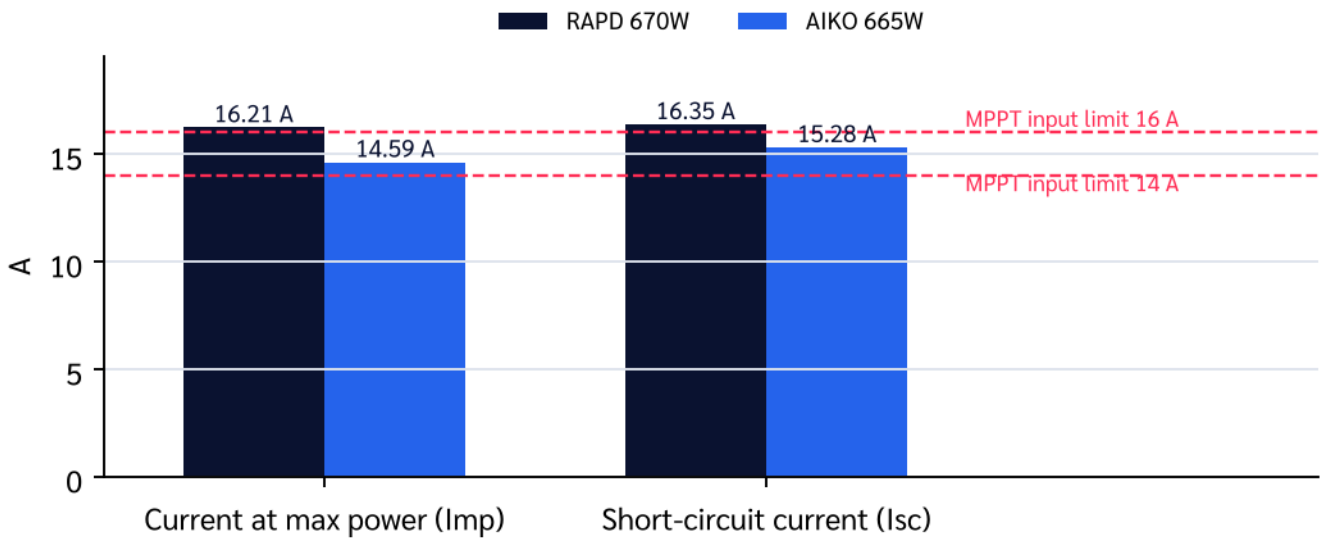
In a back-contact cell, both the positive and negative contacts sit on the back, so nothing on the front blocks light. That gives higher efficiency, an all-dark appearance and, in independent tests, slightly lower losses from heat, low-angle sun and partial shade. "ABC" (All Back Contact) is AIKO's name for its version of the technology. "BC" is the general term, used by LONGi (HPBC) and others.

2 Specifications side by side

Item	RAPD 670W	AIKO 665W	Source / note
Maximum power (Pmax)	670 W	665 W	RAPD: listing · AIKO: datasheet
Module efficiency	24.8%	24.6%	Pmax ÷ (2.382 × 1.134 m²)
Voltage at Pmax (Vmp)	41.35 V	45.60 V	
Current at Pmax (Imp)	16.21 A	14.59 A	
Open-circuit voltage (Voc)	50.17 V	54.35 V	
Short-circuit current (Isc)	16.35 A	15.28 A	
Power temperature coefficient	not published	−0.26 %/°C	BC panels typically −0.26 %/°C
Voc temperature coefficient	not published	−0.22 %/°C	
Faces that generate	2 (bifacial, claimed)	1 (Comet 3N72 Mw, single glass)	AIKO also sells a dual-glass bifacial 665 W (Stellar, 33 kg)
Size	about 2,382 × 1,134 mm	2,382 × 1,134 × 30 mm	
Weight	about 28–32 kg	28.2 kg	
Product warranty	about 12 years (depends on seller)	15 years (datasheet)	Thai dealers may quote 12 years: check who honours it
Power warranty	30 years	30 years, ≥ 88.85% at year 30	year 1 ≤ 1%, then ≤ 0.35%/year
Max series fuse	not published	25 A	
Manufacturer standing	Thai brand; maker not disclosed	BloombergNEF Tier 1 (2026)	
Where to buy in Thailand	online (Lazada, Shopee) and solar shops	authorised distributors and dealers	

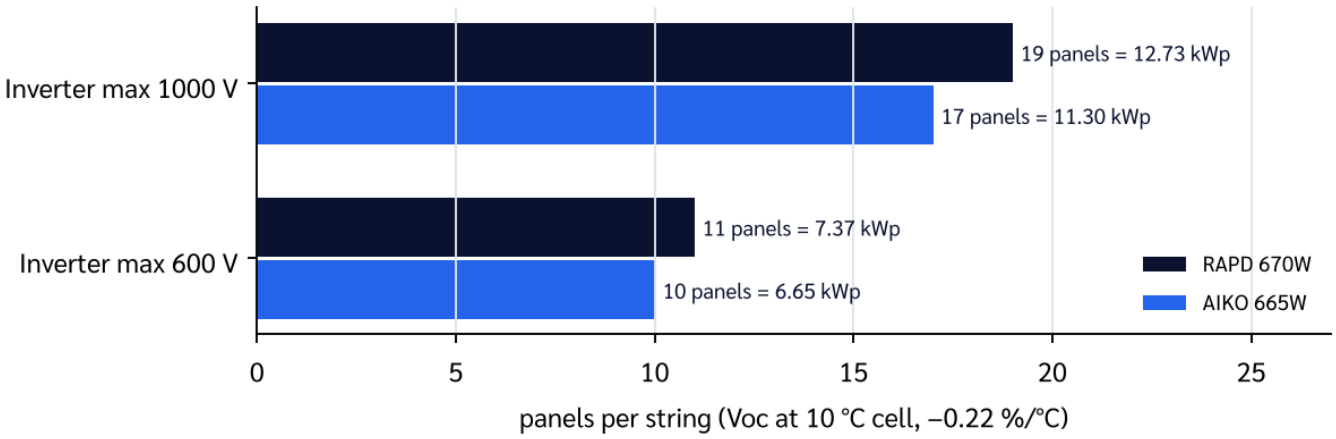
Verification note. The RAPD 670 W BC figures are from the seller's listing. The RAPD panels we found on Thai retail sites (450–750 W) are TOPCon, HJT and half-cell N-type, and we could not find a published RAPD back-contact datasheet. Its electrical values do not match LONGi's 670 W BC panel either (54.3 V / 15.6 A), so the cell supplier is unknown. Treat these numbers as unconfirmed until you have the datasheet, IEC 61215 / 61730 certificates and the flash-test report for the actual batch.

3 Matching the inverter: current is the key difference



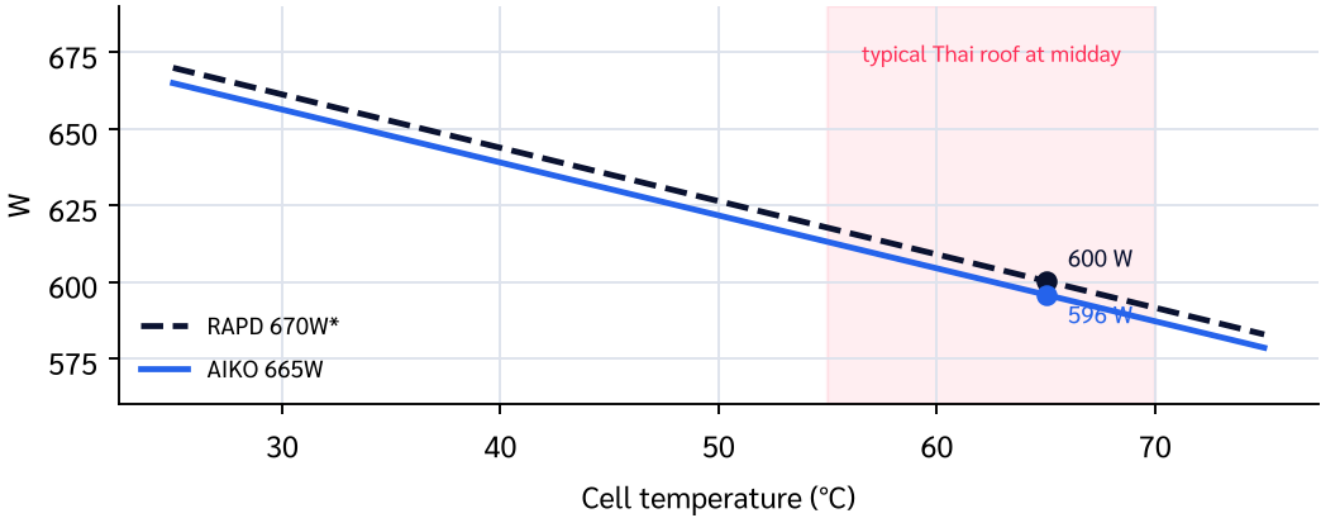
Many inverters for homes and small businesses accept 14–16 A per MPPT input; newer models accept 20–32 A. **RAPD (16.21 A) exceeds a 16 A input slightly, so the inverter clips up to about 1% at peak sun, and it exceeds a 14 A input by about 15%.** AIKO (14.59 A) fits a 16 A input with room to spare. Isc must also stay within the inverter's maximum short-circuit input current. With two strings in parallel on one MPPT, double these figures.

4 String length: RAPD fits more panels per string



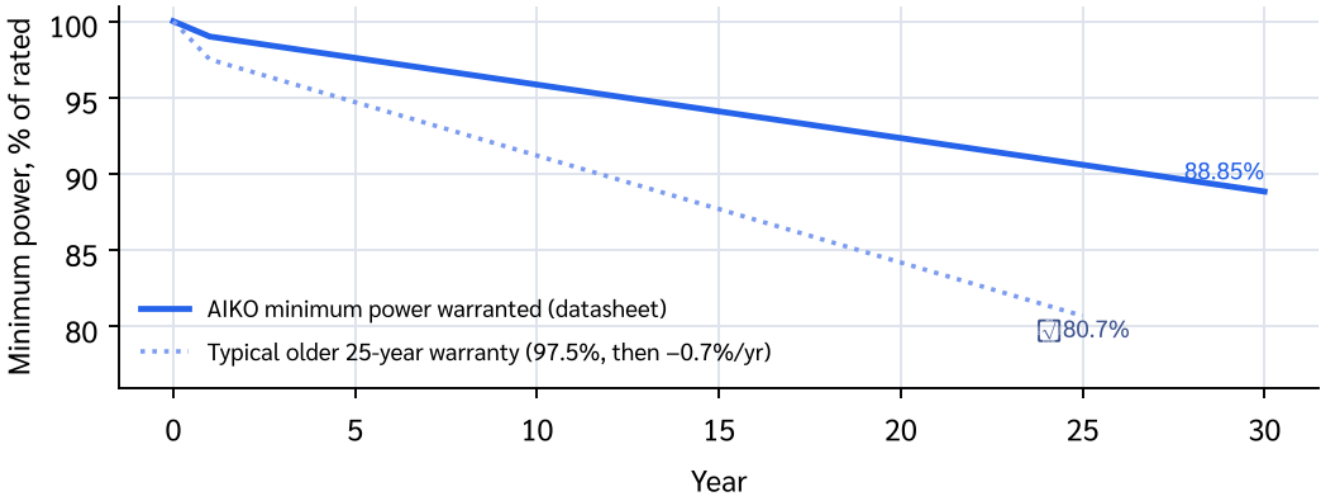
RAPD's lower voltage lets one more panel go on each string on a 600 V inverter (11 against 10), and two more on 1,000 V (19 against 17). That can save a string or an MPPT input on larger systems. Check the low end too: in the heat (65 °C cell), Vmp falls to about 37.7 V (RAPD) and 41.6 V (AIKO) per panel, and the string must stay above the inverter's minimum MPPT voltage. Use the coldest morning at your site; northern Thailand can reach 5–10 °C in winter.

5 Heat: Thai rooftops run hot



At a 65 °C cell, both lose about 10% of rated power: AIKO about 596 W and RAPD about 600 W, assuming the same –0.26 %/°C (*RAPD has not published its coefficient). For comparison, TOPCon panels are typically –0.29 to –0.30 %/°C and PERC –0.34 to –0.35 %/°C, so either back-contact panel keeps about 1–3% more power on a hot roof.

6 Warranty and degradation



AIKO warrants at least 99% after year 1 and loses no more than 0.35% a year after that, ending at 88.85% after 30 years. RAPD offers a 30-year power warranty, but we have not seen its degradation curve. Ask for it in writing, and ask who in Thailand handles claims. A 15-year product warranty from an established maker is worth more than a longer one that nobody in the country can honour.

7 Which one suits your project?

RAPD 670W BC suits

- Inverters with 20 A or more per MPPT input
- Larger strings: one or two more panels per string
- Ground or elevated mounts over light-coloured surfaces, where the bifacial rear side adds yield
- Buyers who need fast online delivery and can verify the datasheet and warranty themselves

AIKO 665W ABC suits

- Inverters with 14–16 A MPPT inputs (common in homes)
- Rooftops where a known datasheet, a 15-year product warranty and Tier 1 standing matter
- Roofs with some shade from trees, poles or neighbouring buildings
- Projects that need documents for PEA, banks or insurers

8 Checklist before buying either panel

1. Get the official datasheet for the exact model and wattage, and keep it with the project file.
2. Ask for IEC 61215 / IEC 61730 certificates and, for Thailand, the TISI (มอก.) listing.
3. Check Imp and Isc against the inverter's maximum MPPT input current and maximum short-circuit current.
4. Work out string length from Voc on the coldest morning and Vmp on the hottest afternoon.
5. Get the warranty terms in writing: product years, power curve, and the Thai company that handles claims.
6. Ask for flash-test results for the delivered batch and check a few panels on arrival.

Our view: both are modern back-contact panels with almost identical power and efficiency. The real differences are electrical (RAPD's 16.2 A current), documentation and warranty. For most Thai homes with 14–16 A inverter inputs, AIKO is the easier and better-documented fit. RAPD can make sense with high-current inverters and verified paperwork.

References

1. AIKO Comet 3N72 AIKO-A665-MDE72Mw — specifications (Voc, Isc, Vmp, Imp, temperature coefficients, NOCT, fuse, warranty)
https://solarproof.com.au/products/AIKO-A665-MDE72MwComet3N72.php
2. Solar-Stack — AIKO Comet 3N72 series data (28.2 kg, 144 cells, –0.26 %/°C)
https://www.solar-stack.com/en/panel/aiko-solar/comet-comet-3n72
3. QES Energy (Thai distributor) — AIKO ABC explained: efficiency, degradation 1% then 0.35%/yr, 88.85% at year 30
https://www.qesenergy.co.th/en/articles/why-aiko-solar-panels
4. ComparePV — AIKO-A665-MDE72Dw (dual-glass bifacial variant, 33 kg)
https://comparepv.com/panel/aiko-aiko-a665-mde72dw
5. BayWa r.e. — LONGi, Jinko, Trina, AIKO and JA Solar confirmed as BloombergNEF Tier 1 in 2026
https://solar-distribution.baywa-re.es/en/about-us/news/details/longi-jinko-trina-aiko-y-ja-solar-se-consolidan-como-fabricantes-de-modulos-tier-1-en-2026
6. LONGi Hi-MO X10 Scientist LR7-72HVH 670M datasheet — a reference BC module at 670 W (–0.26 %/°C, 15/30-year warranty)
https://static.longi.com/LR_7_72_HVH_655_670_M_30_30_and_15_Scientist_BGV_02_20241118_EN_Draft_9293e45f42.pdf
7. TaiyangNews — TÜV Rheinland LCOE comparison of TOPCon and back-contact modules
https://taiyangnews.info/technology/lcoe-based-comparison-of-topcon-and-back-contact-modules
8. RAPD panel listings in Thailand (Lazada, R3 Solarcell, Sunnergytech) — models and technologies on sale
https://www.r3solarcell.com/category/20/ · https://www.sunnergytech.com/category/470/

Prepared by Solar ESS for general information, 30 October 2026. Specifications change between batches and markets; always confirm against the manufacturer's datasheet for the exact model. RAPD values are from the seller's listing and have not been verified against a manufacturer datasheet. String-length and heat figures are worked examples, not a design.